

$\mathbb{E}_\infty$ -DESCENDABILITY FOR MORAVA E -THEORY

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ABSTRACT. Let E_n be the Morava theory of the Honda formal group over $\mathbf{F}_{p^n}$. We prove that $E_n^{hG} \rightarrow E_n$ is $\mathbb{E}_\infty$ -descendable for every closed subgroup G of the extended Morava stabilizer group, both in $K(n)$ -local spectra and in ordinary spectra. In particular, this holds for $L_{K(n)}\mathbb{S} \rightarrow E_n$. For groups of finite p -cohomological dimension, the algebra retraction comes from a section after relative p -finite completion of a finite join. Finite cyclic quotients are treated by a unital arity obstruction tower, whose possible obstructions are restricted to the arities $p, p^2, \dots, p^n$ by chromatic vanishing for partition spectra.

1. INTRODUCTION

A map $A \rightarrow B$ of commutative ring spectra is descendable if the unit of some finite Amitsur stage admits an A -module retraction. Antieau and Stefanich call the map $\mathbb{E}_\infty$ -descendable when such a retraction can be chosen in commutative A -algebras [1, Definition 2.1 and Proposition 2.3]. We prove this stronger form of descent for continuous homotopy fixed points of Morava E -theory.

Fix a prime p and a height $n \geq 1$. Let E_n be the Morava theory of the Honda formal group over $\mathbf{F}_{p^n}$, so that

$$\pi_* E_n = W(\mathbf{F}_{p^n})[[u_1, \dots, u_{n-1}]]\langle u^{\pm 1} \rangle, \quad |u| = -2.$$

Let $\mathcal{O}_n$ be the maximal order in the division algebra over $\mathbf{Q}_p$ of invariant $1/n$, and let

$$\mathbb{G}_n = \mathcal{O}_n^\times \rtimes \mathrm{Gal}(\mathbf{F}_{p^n}/\mathbf{F}_p)$$

be the extended Morava stabilizer group. For a closed subgroup $G \leq \mathbb{G}_n$, we write E_n^{hG} for the continuous homotopy fixed points of Devinatz–Hopkins. The residue field is $\mathbf{F}_{p^n}$ throughout.

Li and Zhang prove that $E_n^{hG} \rightarrow E_n$ is descendable in $K(n)$ -local spectra [10, Corollary 3.2.10]. Our main result is the following.

Theorem 1.1. *For every closed subgroup $G \leq \mathbb{G}_n$, the map*

$$E_n^{hG} \longrightarrow E_n$$

is $\mathbb{E}_\infty$ -descendable in $\mathrm{Sp}_{K(n)}$ and in Sp .

The identification $E_n^{h\mathbb{G}_n} \simeq L_{K(n)}\mathbb{S}$ [10, Theorem 3.1.11] gives the following special case.

Corollary 1.2. *The unit map $L_{K(n)}\mathbb{S} \rightarrow E_n$ is $\mathbb{E}_\infty$ -descendable.*

The proof reduces to two cases. Every closed $G \leq \mathbb{G}_n$ has an open normal uniform pro- p subgroup U with finite solvable quotient. A composition series of G/U therefore reduces the theorem to the map $E_n^{hU} \rightarrow E_n$ and to finite cyclic Galois extensions between continuous homotopy fixed points. These reductions are carried out in Section 7.

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For groups of finite p -cohomological dimension, Proposition 3.1 constructs an algebra retraction geometrically. A finite fiberwise join over BG has cochains in the finite-limit closure of algebras receiving E_n . Its relative p -finite completion admits a section and has the same Morava cochains. Pullback along the section gives the required retraction.

For finite cyclic extensions, ordinary descendability supplies a module retraction, but not the coherent multiplicative homotopies required of an algebra retraction. Theorem 6.1 constructs these homotopies by changing the source. A nilpotent Euler class gives both the initial module retraction and restriction maps between representation-sphere cochains that are null on unit cofibers. Precomposition with such a restriction kills the next unital obstruction by Lemma 4.1. Lemma 5.1 confines the possible obstructions to $p, p^2, \dots, p^n$. After finitely many changes of source, the source stays fixed and all remaining extension spaces are contractible. Lemma 4.3 isolates this argument; no multiplicative lift on the original source is needed.

The obstruction theory records homotopies for every insertion of the unit. This condition cannot be replaced by asking for a partial nonunital algebra map under the scalar algebra: Appendix A gives a counterexample. The result asserts the existence of a finite Amitsur retraction; it does not determine the least possible stage.

Conventions. All limits, colimits, tensor products, and mapping spaces are derived. We write $\mathrm{Map}_A(M, N)$ for the mapping space of A -modules. In the $K(n)$ -local category, relative tensor products and homotopy orbits are formed using the local monoidal structure. Tensoring a spectrum with a local module includes localization. The unit cofiber of a commutative algebra is used only as a module; no augmentation on it is assumed.

2. FINITE LIMITS AND LOCALIZATION

Definition 2.1. A map $A \rightarrow B$ of commutative algebras in a presentable stable symmetric monoidal category is $\mathbb{E}_\infty$ -descendable if, for some $m \geq 0$, the unit

$$A \longrightarrow \mathrm{Tot}^m(B^{\otimes_A(\bullet+1)})$$

admits a retraction in commutative A -algebras.

We will use the equivalent finite-limit criterion of [1, Proposition 2.3]: A must belong to the closure under finite limits and retracts of the full subcategory of commutative A -algebras admitting a map from B . The maps in such a finite diagram need only be A -algebra maps; the choices of maps from B need not be compatible. We also use closure under composition [1, Lemma 2.4(a)].

Lemma 2.2. *Let $A \rightarrow B$ be a map of $K(n)$ -local commutative ring spectra. It is $\mathbb{E}_\infty$ -descendable in ordinary spectra if and only if it is $\mathbb{E}_\infty$ -descendable in $K(n)$ -local spectra.*

Proof. The localization functor $\mathrm{Sp} \rightarrow \mathrm{Sp}_{K(n)}$ is exact and symmetric monoidal. Since partial totalization is a finite-limit construction, it gives an equivalence

$$L_{K(n)} \mathrm{Tot}^m(B^{\otimes_A(\bullet+1)}) \simeq \mathrm{Tot}^m(B^{\otimes_{A, K(n)}(\bullet+1)}).$$

Localizing an ordinary algebra retraction proves one direction. For the converse, precompose a local algebra retraction with the localization map from the ordinary partial totalization. The lax symmetric monoidal inclusion $\mathrm{Sp}_{K(n)} \hookrightarrow \mathrm{Sp}$ regards this composite as a retraction of ordinary commutative A -algebras. $\square$

3. SUBGROUPS OF FINITE COHOMOLOGICAL DIMENSION

Proposition 3.1. *If $G \leq \mathbb{G}_n$ is closed and $\mathrm{cd}_p(G) < \infty$, then $E_n^{hG} \rightarrow E_n$ is $\mathbb{E}_\infty$ -descendable in ordinary spectra.*

The finite join over BG gives the finite-limit expression required by Section 2, but it need not give a section. Relative p -finite completion will supply a section, at the cost of leaving the finite construction. We therefore prove two assertions: the completed join has a section (Lemma 3.2), and completion preserves its Morava cochains (Lemma 3.3). Together they give a retraction through an algebra with the required finite-limit expression.

We work in light condensed anima. Write $\mathrm{CondAn}^{\mathrm{light}}$ for this infinity-topos and $\mathrm{ProfAn}^{\mathrm{light}}$ for its full subcategory of light profinite anima. For a light profinite group G , its classifying object satisfies

$$BG \simeq \lim_i B(G/G_i) \in \mathrm{ProfAn}^{\mathrm{light}},$$

where (G_i) is a countable cofinal system of open normal subgroups [4, Example 4.32].

An anima is p -finite if it is truncated, has finitely many components, and has finite p -groups as its positive homotopy groups. A map in $\mathrm{CondAn}^{\mathrm{light}}$ is p -finite étale if it is locally a projection with p -finite fiber. Denote by $\mathcal{C}_p(BG)$ the full subcategory of these objects in $\mathrm{CondAn}_{/BG}^{\mathrm{light}}$. It is closed under finite limits.

Lemma 3.2. *Let G be a compact p -adic analytic group with $d = \mathrm{cd}_p(G) < \infty$, and choose $m \geq \max\{2, d\}$. Form the fiberwise join of $m + 1$ copies of $* \rightarrow BG$:*

$$(3.1) \quad \mathrm{Join}_m(BG) = \mathrm{colim}_{\emptyset \neq S \subseteq \{0, \dots, m\}} \prod_{s \in S, BG} *,$$

where the subsets are ordered by reverse inclusion. This join has a relative p -finite completion

$$c : \mathrm{Join}_m(BG) \longrightarrow \widehat{\mathrm{Join}}_m^p(BG)$$

over BG with the following properties:

- (1) $\widehat{\mathrm{Join}}_m^p(BG)$ lies in $\mathrm{ProfAn}_{/BG}^{\mathrm{light}}$;
- (2) c induces an equivalence on mapping spaces into every object of $\mathcal{C}_p(BG)$;
- (3) $\widehat{\mathrm{Join}}_m^p(BG) \rightarrow BG$ admits a section.

Proof. We construct the completion as a countable inverse limit of p -finite étale objects over BG . Connectivity and cohomological dimension give a section of each finite object. Finiteness of the sets of section components then gives a section of their inverse limit.

The relative completion. Pullback along $* \rightarrow BG$ identifies the fiber of (3.1) with $G^{*(m+1)}$. The connectivity theorem for joins implies that $\mathrm{Join}_m(BG) \rightarrow BG$ is m -connective. Here an m -connective map has $(m - 1)$ -connected fibers. It therefore induces an equivalence on maps into every $(m - 1)$ -truncated object of the slice.

Consider the infinity-category of factorizations $\mathrm{Join}_m(BG) \rightarrow Z \rightarrow BG$ with $Z \in \mathcal{C}_p(BG)$. This category is cofiltered, since finite limits of the targets supply the required cones. We claim that the resulting pro-object admits a countable presentation.

By [4, Lemmas 4.22 and 4.26(3)], every π -finite étale object over BG descends to a finite quotient stage of BG . There are countably many such stages and countably many π -finite homotopy types and maps. It remains to count the data of a map from $\mathrm{Join}_m(BG)$. A vertex $\prod_{s \in S, BG} * \simeq G^{|S|-1}$ of (3.1) maps to $*$ over BG , trivializing the pullback of Z . Its mapping space into Z is consequently the space of continuous maps from a second-countable profinite set into a π -finite anima. This is a countable filtered colimit of mapping spaces from finite quotients, and hence has a countable simplicial model. Taking the finite limit prescribed by (3.1) gives the

same conclusion for $\mathrm{Map}_{BG}(\mathrm{Join}_m(BG), Z)$. Thus the factorization category has a countable simplicial model. Applying [11, Corollary 9.1.8.10, Tag 0628] to its opposite gives a cointial tower $Z_0 \leftarrow Z_1 \leftarrow \cdots$.

Clausen identifies $\mathrm{ProfAn}_{/BG}^{\mathrm{light}}$ with the countable pro-category of π -finite étale objects over BG [4, Lemma 4.26(4)]. We may therefore realize this tower in the slice and set $\widehat{\mathrm{Join}}_m^p(BG) = \lim_i Z_i$. Pro-Yoneda and this fully faithful realization give

$$(3.2) \quad \mathrm{Map}_{BG}(\widehat{\mathrm{Join}}_m^p(BG), Z) \xrightarrow{\sim} \mathrm{Map}_{BG}(\mathrm{Join}_m(BG), Z), \quad Z \in \mathcal{C}_p(BG).$$

Sections of the finite factors. The given map $\mathrm{Join}_m(BG) \rightarrow Z_i$ descends, by connectivity, to a section of the relative $(m-1)$ -truncation of $Z_i \rightarrow BG$. The assumption $m \geq 2$ puts all remaining relative homotopy groups in the abelian range. For $q \geq m$, the obstruction to lifting a section from Postnikov degree $q-1$ to degree q belongs to

$$H_c^{q+1}(G; M_q),$$

where M_q is the finite p -primary local system obtained by pulling back the relative homotopy group along the section. This group vanishes because $q+1 > d$. The relative Postnikov tower is finite, so every $Z_i \rightarrow BG$ admits a section.

Each of these section spaces also has finitely many components. For the cohomological input, choose an open normal uniform pro- p subgroup. Lazard's calculation gives finite cohomology with $\mathbf{F}_p$ coefficients in every degree [4, Section 5, before Definition 5.14]. A filtration of a finite p -primary module, followed by the Hochschild–Serre spectral sequence for the finite quotient, gives finiteness for G . At Postnikov degree one, nonabelian H^1 is finite because a continuous crossed homomorphism into a finite group is determined by finitely many topological generators. The degree-zero choices are finite as well. Induction through the Postnikov tower now gives the asserted finiteness of section components.

A section of the inverse limit. The component sets form a tower of nonempty finite sets, whose inverse limit is nonempty. Choose section points in compatible components and paths connecting successive images. Such points and paths determine a point in the homotopy limit of a tower of spaces. Therefore

$$\mathrm{Map}_{BG}(BG, \widehat{\mathrm{Join}}_m^p(BG)) \simeq \lim_i \mathrm{Map}_{BG}(BG, Z_i) \neq \emptyset.$$

This proves the third assertion. $\square$

Now let $G \leq \mathbb{G}_n$ be closed, still with $\mathrm{cd}_p(G) < \infty$. Let $\mathcal{E}_n$ be Clausen's continuous Morava E -theory sheaf restricted to BG . It is a commutative algebra in spectrum objects of $\mathrm{CondAn}_{/BG}^{\mathrm{light}}$. Its sections satisfy

$$(3.3) \quad \Gamma(BG; \mathcal{E}_n) \simeq E_n^{hG}, \quad \Gamma(*; \mathcal{E}_n) \simeq E_n$$

by [4, Theorem 12.15 and Remark 12.16].

The mapping property (3.2) applies only to p -finite targets. To apply it to $\mathcal{E}_n$, we pass to Moore quotients and then to finite Postnikov truncations.

Lemma 3.3. *The completion map in Lemma 3.2 induces an equivalence of commutative E_n^{hG} -algebras*

$$\Gamma(\widehat{\mathrm{Join}}_m^p(BG); \mathcal{E}_n) \xrightarrow{\sim} \Gamma(\mathrm{Join}_m(BG); \mathcal{E}_n).$$

Proof. Choose a generalized Moore tower (M_j) computing completion at $I_n = (p, v_1, \dots, v_{n-1})$. We have

$$(3.4) \quad \mathcal{E}_n \simeq \lim_j (\mathcal{E}_n \wedge M_j).$$

Indeed, on its defining profinite torsor charts $\mathcal{E}_n$ takes values in the complete generalized Morava theories of [4, Definitions 12.6 and 12.9]. Smash product with a finite spectrum is computed objectwise, which proves (3.4) also with the continuous action.

After pullback to $*$, the sheaf $\mathcal{E}_n \wedge M_j$ is constant with value $E_n \wedge M_j$. Clausen identifies the underlying condensed $\mathcal{E}_n$ with the I_n -completion of the constant E_n [4, Lemma 12.10]; taking a generalized Moore quotient of that completion agrees with taking the quotient before completion. The homotopy groups of $E_n \wedge M_j$ are finite p -primary groups. Thus $\tau_{\leq r} \Omega^\infty \Sigma^k(\mathcal{E}_n \wedge M_j)$ is a p -finite étale object over BG for every $k \in \mathbf{Z}$ and $r \geq 0$, and (3.2) applies to it.

Every slice of $\text{CondAn}^{\text{light}}$ is Postnikov complete [4, Theorem 4.11]. Taking the limit over r and then varying k therefore gives an equivalence of section spectra with coefficients in $\mathcal{E}_n \wedge M_j$. Taking the limit over j and using (3.4) proves the claim on underlying spectra. The map is a map of commutative algebras, so it is an equivalence there as well. No commutative algebra structures on the Moore quotients were used. $\square$

Proof of Proposition 3.1. Choose a section s as in Lemma 3.2. Pullback on cochains gives a retraction

$$E_n^{hG} \longrightarrow \Gamma(\widehat{\text{Join}}_m^p(BG); \mathcal{E}_n) \xrightarrow{s^*} E_n^{hG}.$$

By Lemma 3.3, this is a retraction through $\Gamma(\text{Join}_m(BG); \mathcal{E}_n)$. The finite colimit (3.1) expresses that algebra as a finite limit of algebras $\Gamma(\prod_{s \in S, BG} s; \mathcal{E}_n)$. Each receives E_n from a projection to $*$, and every map in the diagram is an E_n^{hG} -algebra map. The finite-limit criterion proves the proposition. $\square$

Remark 3.4. The section argument uses cohomological dimension only with finite discrete coefficients. No identification of the fiber after relative completion is needed. The coefficient argument uses Postnikov completeness rather than a spectral sequence with unbounded Morava coefficients. In particular, the proposition applies to uniform pro- p groups by Lazard's theorem, and to compact p -adic analytic groups without elements of order p by [13, Corollary (1)].

4. UNITAL ARITY OBSTRUCTIONS

The binary case explains why every unit insertion must enter the obstruction theory. Let X be a commutative A -algebra with unit $u : A \rightarrow X$, and let $q : X \rightarrow A$ be an A -module retraction with a specified homotopy $qu \simeq \text{id}_A$. Write $M = \text{cofib}(u)$. The binary multiplicative defect is

$$q\mu_X - \mu_A(q \otimes_A q) : X \otimes_A X \longrightarrow A.$$

A unital multiplicativity homotopy must restrict on $A \otimes_A X$ and $X \otimes_A A$ to the homotopies supplied by $qu \simeq \text{id}_A$ and the unit laws. These restrictions agree coherently on $A \otimes_A A$. Thus the prescribed boundary is the unit fat wedge

$$(4.1) \quad \begin{aligned} F_2(u) &= (A \otimes_A X) \amalg_{A \otimes_A A} (X \otimes_A A), \\ \text{cofib}(F_2(u) \rightarrow X \otimes_A X) &\simeq M \otimes_A M. \end{aligned}$$

Fixing only the all-unit input omits the two mixed-input conditions. Appendix A shows why this omission matters. The general arity calculation below combines the full unit fat wedge with the boundary prescribed by lower positive compositions.

Let Comm be the positive commutative operad in spectra, with $\text{Comm}(0) = 0$ and $\text{Comm}(a) = \mathbb{S}$ for $a \geq 1$. Let φ_{a-1} denote derived operadic extension from arities at most $a-1$, and define the Σ_a -spectrum

$$(4.2) \quad C_a = \text{cofib}((\varphi_{a-1} \text{Comm})(a) \longrightarrow \text{Comm}(a)), \quad a \geq 2.$$

The following lemma uses these positive-arity coefficients to obstruct *unital* algebra retractions. Its tower records compatibility with every insertion of a unit.

Lemma 4.1. *Let X be a commutative A -algebra, in ordinary or $K(n)$ -local spectra, with unit $u : A \rightarrow X$ and unit cofiber $M = \text{cofib}(u)$. For $a \geq 2$, put*

$$Q_a(M) = (C_a \otimes M^{\otimes A^a})_{h\Sigma_a}.$$

There is a natural tower of spaces $\text{Ret}_{\leq a}(X/A)$, for $a \geq 1$, with the following properties.

- (1) *The first stage consists of A -module maps $q : X \rightarrow A$ with a specified homotopy $qu \simeq \text{id}_A$.*
- (2) *The inverse limit is $\text{Map}_{\text{CAlg}_A}(X, A)$.*
- (3) *A point q_{a-1} of $\text{Ret}_{\leq a-1}(X/A)$ has an obstruction*

$$(4.3) \quad \kappa_a(q_{a-1}) \in \pi_0 \text{Map}_A(Q_a(M), A).$$

It extends to $\text{Ret}_{\leq a}(X/A)$ if and only if this class is zero. A nullhomotopy gives an extension. When nonempty, the extension space is a torsor for $\Omega \text{Map}_A(Q_a(M), A)$.

- (4) *For a commutative A -algebra map $f : X' \rightarrow X$, precomposition is defined at every stage and*

$$(4.4) \quad \kappa_a(q_{a-1}f) = \kappa_a(q_{a-1}) \circ Q_a(\bar{f}),$$

where $\bar{f} : \text{cofib}(A \rightarrow X') \rightarrow M$ is the induced map. If $\bar{f}$ is null as a module map, the precomposed point extends through arity a .

Proof. We first construct a unitary operad resolution with arity one fixed. Its truncated morphism spaces give the tower and its first stage. In arity a , the lower operations and the unit insertions prescribe a common boundary. We identify the cofiber of this boundary with $C_a \otimes M^{\otimes A^a}$; the extension problem and its naturality then give (3) and (4).

A unitary resolution. A unitary operad has a single nullary operation. Equivalently, its positive part is a Λ -operad: besides permutations and positive compositions, it has restriction maps along injections of finite sets, obtained by inserting the nullary operation in the omitted inputs. Restriction remembers each unit insertion, not just the input consisting entirely of units.

We use the model structure on truncated Λ -operads from [7, Section 10.1]. Its cofibrations are exactly the cofibrations of underlying positive operads [7, Proposition 10.1]. Restriction of arity has a left Quillen adjoint; the arity extensions of a cofibrant operad form a cofibrant exhaustive filtration [7, Propositions 10.2–10.3 and Construction 10.4]. Comparison with derived mapping spaces of ordinary unitary operads, also after truncation, is given by [6, Theorems 1 and 1'].

Choose a cofibrant unitary resolution $P \rightarrow \text{Comm}^u$ in spaces with $P(0) = P(1) = *$. Here Comm^u is the unitary commutative operad in spaces. A choice keeping arity one fixed can be constructed inductively. Given P in smaller arities, put

$$L_a P = (\text{Ext}_a(P_{<a}))(a), \quad \text{Match}_a P = \lim_{S \subsetneq [a]} P(S),$$

where Ext_a is left extension to arity a and $[a] = \{1, \dots, a\}$. The latching object consists of decomposable positive operations; the matching object records unit insertions. Factor the pruning map in Σ_a -spaces as a cofibration followed by a trivial fibration:

$$L_a P \hookrightarrow P(a) \xrightarrow{\sim} \text{Match}_a P.$$

These maps specify all new compositions and restrictions. A nonidentity positive composition of output arity a either uses only smaller arities or consists of a single a -ary operation and unary identities. Nullary compositions are the specified restrictions; compatibility on $L_a P$ is exactly their required compatibility with positive composition.

Start with $P(0) = P(1) = *$. Inductively, the proper-subset diagram is finite, Reedy fibrant, and objectwise contractible, so its limit and $P(a)$ are contractible. The latching cofibration gives positive operadic cofibrancy, and the matching fibration gives the next Reedy fibrancy condition. This constructs the claimed resolution without changing unary operations.

Forgetting restrictions identifies the positive part of $\text{Ext}_a(P_{<a})$ with the usual positive free arity extension. On trees, prune nullary inputs, contract unary identities, and apply the already specified restrictions at each vertex. The unitary axioms make this independent of the order of pruning and compatible with contracting internal edges. This defines all restriction maps on the positive free extension. Given a map from $P_{<a}$ to any truncated Λ -operad, its unique positive extension commutes with these restrictions: this holds at each vertex by hypothesis and therefore on every pruned tree. The positive extension, with these restriction maps, consequently satisfies the universal property of Ext_a . Cofibrancy makes this a derived identification. Applying Σ_+^∞ therefore gives

$$(4.5) \quad \text{cofib}(\Sigma_+^\infty L_a P \longrightarrow \Sigma_+^\infty P(a)) \simeq C_a$$

equivariantly. Here the coefficient is the one in (4.2); its relation to the positive operadic bar construction is [9, Remark 4.3].

The tower of retractions. Work in $(\text{Mod}_A)_{A/}$ with tensor product induced by $\otimes_A$; its tensor unit id_A is initial. For $a \geq 1$, let

$$P^{[a]} = \mathbb{L} \text{Ext}_\infty^\Lambda(P_{\leq a})$$

be the derived free extension to a full unitary operad. This extension retains the nullary operation and all Λ -restrictions. Pull back the structures on X and A along $P^{[a]} \rightarrow P \rightarrow \text{Comm}^u$, and define

$$(4.6) \quad \text{Ret}_{\leq a}(X/A) = \text{Map}_{\text{Alg}_{P^{[a]}}((\text{Mod}_A)_{A/})}(X, A).$$

The maps $P^{[a-1]} \rightarrow P^{[a]}$ give the restriction tower. The operad $P^{[1]}$ has only the identity and the nullary unit, so its algebra category is $(\text{Mod}_A)_{A/}$. This gives (1). The exhaustive filtration $\text{colim}_a P^{[a]} \simeq P$ [7, Proposition 10.3 and equation (34)] gives (2). Precomposition with a commutative A -algebra map is defined at every stage of (4.6).

To compute the fibers, fix a pointed module map $q : X \rightarrow A$. Write $\text{End}(q)$ for its endomorphism operad in the pointwise monoidal arrow category, and $\text{End}(X)$ and $\text{End}(A)$ for the endpoint endomorphism operads in $(\text{Mod}_A)_{A/}$. Let $\text{Map}_{A/}$ denote pointed module mapping spaces. The arrow-category mapping formula gives

$$(4.7) \quad \text{End}(q)(a) \simeq \text{Map}_{A/}(X^{\otimes A^a}, X) \times_{\text{Map}_{A/}(X^{\otimes A^a}, A)}^h \text{Map}_{A/}(A^{\otimes A^a}, A).$$

The two maps to the middle space are postcomposition with q and precomposition with $q^{\otimes A^a}$. Hence fixing the endpoint operations leaves the space of homotopies making their square commute. This identity respects positive composition and every unit insertion.

Algebras in a pointwise monoidal arrow category are arrows of algebras. Consequently, the fiber of $\text{Ret}_{\leq a}(X/A) \rightarrow \text{Map}_{A/}(X, A)$ over q is the space of $P^{[a]}$ -structures on this arrow with the endpoint structures fixed. By the free-extension adjunction, it is equally the corresponding derived space of $P_{\leq a}$ -maps into $\text{End}(q)$. We use this description only to calculate the extension fiber.

The relative matching square. For a unitary operad O , write $\text{Match}_a O = \lim_{S \subsetneq [a]} O(S)$. Fix a point q_{a-1} of $\text{Ret}_{\leq a-1}(X/A)$ and its underlying pointed arrow q . Replace the comparison

$$\text{End}(q) \longrightarrow \text{End}(X) \times \text{End}(A)$$

by a Reedy fibration between Reedy-fibrant unitary operads, carrying along the specified endpoint structures and the lower-stage lift. We retain the notation for these compatible models. Their relative matching maps are fibrations; ordinary pullbacks therefore compute the homotopy pullbacks below.

The matching object with the endpoints prescribed is

$$(4.8) \quad \text{Match}_a^\partial(q) = \text{Match}_a \text{End}(q) \underset{\text{Match}_a \text{End}(X) \times \text{Match}_a \text{End}(A)}{\overset{h}{\times}} (\text{End}(X)(a) \times \text{End}(A)(a)).$$

The extension fiber over q_{a-1} is the derived space of Σ_a -equivariant fillers in

$$(4.9) \quad \begin{array}{ccc} L_a P & \longrightarrow & \text{End}(q)(a) \\ \downarrow & \nearrow & \downarrow \\ P(a) & \longrightarrow & \text{Match}_a^\partial(q). \end{array}$$

The top map is prescribed by lower positive compositions. The bottom map has two components: the given arity- a endpoint operations, and the lower-stage operations pulled back along every restriction $P(a) \rightarrow P(S)$ for $S \subsetneq [a]$. They agree after applying the endpoint matching maps because q_{a-1} already has the specified endpoints. Thus the right-hand map imposes both the unit restrictions and the fixed endpoint operations.

The latching–matching description is the one used in the proof of [7, Proposition 10.8]. Although that proof uses the Fulton–MacPherson operad as source, the filler description requires only decomposable latching, identity-only unary operations, and proper-subset matching. Our construction of P supplies these data and the latching cofibrations. The compatible replacement of the comparison arrow supplies the relative matching fibrations. This proves the asserted filler description with the endpoints fixed.

The arity- a boundary. For a fixed pointed arrow q , define the unit fat wedge by

$$(4.10) \quad F_a(u) = \text{colim}_{S \subsetneq [a]} \bigotimes_{i=1}^a X_i(S), \quad X_i(S) = \begin{cases} X & i \in S, \\ A & i \notin S. \end{cases}$$

The subsets are ordered by inclusion and the maps insert u . All cofibers in the remainder of this proof are computed in the stable category Mod_A , not in its undercategory $(\text{Mod}_A)_A/$. Iterating $A \rightarrow X \rightarrow M$ in the independent tensor factors gives an equivariant equivalence

$$(4.11) \quad \text{cofib}(F_a(u) \rightarrow X^{\otimes A^a}) \simeq M^{\otimes A^a}.$$

We now translate (4.9) into a boundary for a homotopy of module maps. By (4.7), the fixed endpoint operations leave a homotopy between $q\mu_X$ and $\mu_A q^{\otimes A^a}$, parametrized by $P(a)$.

The top map in (4.9) prescribes this homotopy on $\Sigma_+^\infty L_a P \otimes X^{\otimes A^a}$. For each proper subset S , the matching component pulls back the lower homotopy along $P(a) \rightarrow P(S)$ and prescribes its restriction to $\Sigma_+^\infty P(a) \otimes \bigotimes_i X_i(S)$. The unit laws of the fixed endpoints identify its two ends with the restrictions of the same arity- a maps. As S varies, these homotopies are coherent and hence give a homotopy on $\Sigma_+^\infty P(a) \otimes F_a(u)$.

Compatibility of positive composition with every Λ -restriction makes the two prescribed homotopies agree on $\Sigma_+^\infty L_a P \otimes F_a(u)$. In particular, $S = \emptyset$ retains the full scalar family $\Sigma_+^\infty P(a) \otimes A$, together with the specified under- A coherences. The homotopy pushout joins exactly these two prescriptions; restricting back to its two terms recovers the latching and matching data.

The resulting boundary map in Mod_A is

$$(4.12) \quad \begin{aligned} & (\Sigma_+^\infty L_a P \otimes X^{\otimes A^a})_{\Sigma_+^\infty L_a P \otimes F_a(u)} \amalg (\Sigma_+^\infty P(a) \otimes F_a(u)) \\ & \longrightarrow \Sigma_+^\infty P(a) \otimes X^{\otimes A^a}. \end{aligned}$$

Tensoring a spectrum with an A -module is implicit here. This is the pushout-product of the positive latching map and the unit fat-wedge map. By (4.5) and (4.11), its cofiber is $C_a \otimes M^{\otimes A^a}$ with its natural permutation action.

For any map $D \rightarrow T$ of A -modules, stability gives

$$\text{Map}_A(\text{cofib}(D \rightarrow T), A) \simeq \text{fib}_0(\text{Map}_A(T, A) \longrightarrow \text{Map}_A(D, A)).$$

If $\alpha, \beta : T \rightarrow A$ are equipped with a homotopy between their restrictions to D , subtracting β gives a point of this fiber: the difference $\alpha - \beta$ together with its prescribed nullhomotopy on D . Extending the original homotopy is equivalent to joining this point to zero in the fiber. This follows by translating the homotopy-lifting problem in the mapping spaces, which are infinite loop spaces.

Apply this description to (4.12), after homotopy orbits, and to the two endpoint maps. Denote the resulting actual map by $\tilde{\kappa}_a(q_{a-1}) : Q_a(M) \rightarrow A$. Its component is the class in (4.3). The preceding identifications give the fiber equivalence

$$(4.13) \quad \begin{aligned} & \text{fib}_{q_{a-1}}(\text{Ret}_{\leq a}(X/A) \longrightarrow \text{Ret}_{\leq a-1}(X/A)) \\ & \simeq \text{Path}_{\text{Map}_A(Q_a(M), A)}(\tilde{\kappa}_a(q_{a-1}), 0). \end{aligned}$$

Here Path means the space of paths with the two indicated endpoints fixed. In particular, no additional suspension occurs. When nonempty, this path space is a torsor for $\Omega \text{Map}_A(Q_a(M), A)$, proving (3).

Naturality. The construction commutes with precomposition by f . The unit cubes give a map of fat wedges, and their cofibers give $\bar{f}^{\otimes A^a}$. The endpoint operations and the specified lower homotopies commute with this precomposition as well. This proves (4.4). The functor Q_a carries a module nullhomotopy of $\bar{f}$ to a nullhomotopy of $Q_a(\bar{f})$; equivariance is built into the tensor-power functor, so no extra equivariant choice is required. $\square$

Remark 4.2. The stages $\text{Ret}_{\leq a}(X/A)$ are not positive $\varphi_a \text{Comm}$ -mapping spaces under A . Those only fix the all-unit restriction and admit the counterexample in Appendix A. Lemma 4.1 requires the unitary latching and matching conditions used above.

Lemma 4.3. *Let $t \geq 0$ and let*

$$X_0 \xleftarrow{f_1} X_1 \longleftarrow \cdots \xleftarrow{f_t} X_t$$

be a chain of commutative A -algebras in ordinary or $K(n)$ -local spectra. Put $M_j = \text{cofib}(A \rightarrow X_j)$. Suppose that $A \rightarrow X_0$ has a pointed A -module retraction q_0 and that every induced map $M_j \rightarrow M_{j-1}$ is null. Suppose also that a set $S \subseteq \{2, 3, \dots\}$ of cardinality t satisfies

$$Q_a(M_j) = 0 \quad (0 \leq j \leq t, \ a \geq 2, \ a \notin S).$$

Then $A \rightarrow X_t$ has a commutative A -algebra retraction whose underlying pointed module map is homotopic to $q_0 f_1 \cdots f_t$.

Proof. Write $S = \{s_1 < \cdots < s_t\}$ and start with q_0 at stage one. Before the j th change of source, the construction has reached arity s_{j-1} on X_{j-1} , with stage one understood when $j = 1$. All intervening arities below s_j lie outside S . Their extension spaces are contractible by Lemma 4.1, so the point extends to arity $s_j - 1$.

Precompose this point with f_j . By (4.4), its arity- s_j obstruction is the preceding obstruction precomposed with $Q_{s_j}(M_j) \rightarrow Q_{s_j}(M_{j-1})$. This map is null, so the precomposed point extends through arity s_j . After the t changes, keep the source X_t fixed. Every remaining extension space is contractible, and the inverse limit in Lemma 4.1 gives an algebra retraction. When $t = 0$, the construction stays on X_0 and uses contractible extensions from the first stage onward. Extensions preserve the underlying pointed module map, and each precomposition composes it with f_j . The final map therefore has the asserted underlying retraction. $\square$

5. PARTITION SPECTRA ON ODD SPHERES

Lemma 5.1. *For every odd integer q and every $a \geq 2$,*

$$L_{K(n)}(C_a \wedge (S^q)^{\wedge a})_{h\Sigma_a} \simeq 0$$

unless $a \in \{p, p^2, \dots, p^n\}$. The action on the sphere factor is its tensor-permutation action, and q may be negative.

There are two distinct vanishing assertions here. Vanishing away from powers of p makes the possible arities sparse; vanishing at p^k for $k > n$ makes the remaining set finite.

Proof. We first identify C_a with a partition spectrum, including the suspension and action. Let $\mathcal{P}(a)_\bullet$ be the pointed simplicial partition set of [3, Section 4]: its nonbasepoint simplices are chains containing both the discrete and indiscrete partitions. Then

$$(5.1) \quad B\text{Comm}(a) \simeq \Sigma^\infty |\mathcal{P}(a)_\bullet|, \quad C_a \simeq \Sigma^{-1} \Sigma^\infty |\mathcal{P}(a)_\bullet|.$$

The first identification is used in the proof of [3, Proposition 4.6]; the second is [9, Remark 4.3]. Equivalently, C_a is the suspension spectrum of the unreduced suspension of the proper partition poset used by Arone–Dwyer. These identifications are Σ_a -equivariant. In particular, $C_2 \simeq \mathbb{S}$, whereas $B\text{Comm}(2) \simeq \Sigma \mathbb{S}$. We use this bar construction, not its Spanier–Whitehead dual.

If a is not a power of p , [2, Theorem 1.8 and Remark 1.11] give mod- p acyclicity of the homotopy-orbit spectrum in the statement. Remark 1.11 includes negative odd suspensions in the stable formulation. This spectrum is bounded below and has finitely generated integral homology in each degree: the partition spectrum and the group Σ_a are finite. Universal coefficients and Nakayama’s lemma imply that its $\mathbf{Z}_{(p)}$ -homology vanishes. Stable Hurewicz, starting at a lower bound, shows that its p -localization is zero. This proves the required $K(n)$ -acyclicity when a is not a power of p .

Suppose now that $a = p^k$ with $k > n$. The calculation needed here is in the *proof* of [3, Lemma 5.6], not only its degree-zero conclusion. Their spectral sequence (5.7) computes completed E_n -homology from the simplicial partition filtration. Let Δ^{-q} be their ring of additive Morava power operations, with grading by weights p^k . Write $\overline{(E_n)_0}$ for the coefficient module on which operations of weight p^j with $j > 0$ act trivially. By Lemmas 5.5 and 3.9 of that paper, for each $t \in \mathbf{Z}$ the normalized complex in internal degree $q + t$ is quasi-isomorphic to the degree- t part of

$$(E_n)_* \otimes_{(E_n)_0} B_\bullet(\overline{(E_n)_0}, \Delta^{-q}, \overline{(E_n)_0})[k],$$

where $[k]$ selects weight p^k . By [3, Theorem 2.20], the weight- p^k bar complex before extension of scalars has homology only in simplicial degree k , given by the module denoted $C[k]_q$ there. This module is zero for $k > n$. Since $(E_n)_*$ is free over $(E_n)_0$, the displayed graded complex is acyclic as well. The spectral sequence therefore vanishes in every internal degree. The normalized partition filtration is finite, of length at most $p^k - 1$, so it follows that

$$L_{K(n)} \left(E_n \wedge (\Sigma^\infty |\mathcal{P}(p^k)_\bullet| \wedge (S^q)^{\wedge p^k})_{h\Sigma_{p^k}} \right) \simeq 0.$$

Completed Morava E -homology detects $K(n)$ -acyclicity: the even-periodic Morava K -theory is an E_n -module, and base change to it detects precisely that Bousfield class. Desuspending once, as in (5.1), proves the assertion. $\square$

In applying this lemma, it is the vanishing of the whole object $Q_a(M)$ that matters. By (4.13), the equality $Q_a(M) = 0$ makes every extension fiber contractible. The weaker condition $\kappa_a = 0$ gives an extension but need not make its choice contractible. Lemma 4.3 uses the former condition to keep the final source fixed while completing the tower.

6. FINITE CYCLIC GALOIS EXTENSIONS

Theorem 6.1. *Let $A \rightarrow B$ be a descendable faithful finite C_ℓ -Galois extension of commutative algebras in $\mathrm{Sp}_{K(n)}$, where ℓ is prime. Then $A \rightarrow B$ is $\mathbb{E}_\infty$ -descendable.*

Proof. We construct a chain of representation-sphere cochain algebras to which Lemma 4.3 applies. Euler-class nilpotence supplies the initial module retraction and the null maps on unit cofibers. Lemma 5.1 supplies the finite set of possible obstructions. The resulting algebra retraction will be on a later cochain algebra: the argument does not require a multiplicative lift on the initial source.

All tensor products in this proof are $K(n)$ -local. Choose a faithful complex character of C_ℓ and let V be its underlying real two-dimensional representation. The sphere $S(rV)$ is free for every $r \geq 1$. The real dimension two will also make the unit cofibers odd suspensions after base change, as (6.2) shows. This parity is needed for Lemma 5.1 at every stage. For $\ell = 2$, we use two copies of the real sign representation.

Euler classes and unit cofibers. For $r \geq 1$, consider the commutative A -algebra

$$\mathrm{Coch}_r(B) = F(S(rV)_+, B)^{hC_\ell}.$$

Galois module descent is a symmetric monoidal equivalence between A -modules and semilinear C_ℓ -equivariant B -modules. Thus the A -module

$$L = (S^{-V} \wedge B)^{hC_\ell}$$

is invertible and satisfies

$$L^{\otimes_{A^r}} \simeq (S^{-rV} \wedge B)^{hC_\ell}, \quad B \otimes_A L \simeq \Sigma^{-2} B.$$

Applying cochains and fixed points to $S(rV)_+ \rightarrow S^0 \rightarrow S^{rV}$ gives an exact sequence

$$(6.1) \quad L^{\otimes_{A^r}} \xrightarrow{e^r} A \longrightarrow \mathrm{Coch}_r(B),$$

where e is the descended Euler class. In particular, the unit cofiber is

$$(6.2) \quad \begin{aligned} M_r &= \mathrm{cofib}(A \rightarrow \mathrm{Coch}_r(B)) \simeq \Sigma L^{\otimes_{A^r}}, \\ B \otimes_A M_r &\simeq \Sigma^{1-2r} B. \end{aligned}$$

The base change of e to B is induced by the map of ordinary sphere spectra $S^{-2} \rightarrow S^0$. This map is null before localization, since $\pi_{-2}\mathbb{S} = 0$ in ordinary spectra; tensoring with B and localizing preserve its nullhomotopy. Apply [12, Proposition 3.27] to a sufficiently long composite

$$L^{\otimes_{A^N}} \longrightarrow L^{\otimes_{A^N}(N-1)} \longrightarrow \cdots \longrightarrow A,$$

each of whose maps tensors e with an identity and becomes null over B . We obtain $N \geq 1$ with $e^N = 0$. The sequence (6.1) gives an A -module retraction of $A \rightarrow \mathrm{Coch}_N(B)$.

Restriction along $S(rV) \subset S((r+N)V)$ gives a commutative A -algebra map $\text{Coch}_{r+N}(B) \rightarrow \text{Coch}_r(B)$. Naturality of (6.1) identifies its map on unit cofibers with

$$(6.3) \quad \Sigma L^{\otimes_A(r+N)} \xrightarrow{\Sigma(e^N \otimes \text{id})} \Sigma L^{\otimes_A r},$$

which is null. We have obtained both the initial module retraction and the null transitions needed to extend it.

The algebra retraction. For the obstruction objects $Q_a(M_r)$ of Lemma 4.1, base change and (6.2) give

$$B \otimes_A Q_a(M_r) \simeq B \otimes (C_a \wedge (S^{1-2r})^{\wedge a})_{h\Sigma_a}.$$

This equivalence respects the tensor-permutation action on the odd sphere factors. Lemma 5.1 and conservativity of descendable base change [12, Proposition 3.19] therefore imply

$$(6.4) \quad Q_a(M_r) = 0 \quad \text{if } a \notin \{p, p^2, \dots, p^n\}.$$

Apply Lemma 4.3 to

$$\text{Coch}_N(B) \longleftarrow \text{Coch}_{2N}(B) \longleftarrow \dots \longleftarrow \text{Coch}_{(n+1)N}(B)$$

and to the set of arities $\{p, p^2, \dots, p^n\}$. The initial pointed module retraction comes from (6.1) and $e^N = 0$. The maps on unit cofibers are null by (6.3), and (6.4) gives the required vanishing. Thus there is a commutative A -algebra retraction

$$(6.5) \quad F(S((n+1)NV)_+, B)^{hC_\ell} \longrightarrow A.$$

Its underlying pointed module map is the initial retraction precomposed with restriction from $\text{Coch}_{(n+1)N}(B)$ to $\text{Coch}_N(B)$.

The finite-limit criterion. For any finite free C_ℓ -CW complex T , the algebra $F(T_+, B)^{hC_\ell}$ lies in the finite-limit closure of the commutative A -algebras receiving B . Indeed, a free cell attachment becomes a pullback of algebras. Shapiro's equivalence identifies the cochains of the attaching cell and its boundary with $F(D_+^j, B)$ and $F(S_+^{j-1}, B)$, both of which receive B by constant functions. The empty complex gives the zero algebra, which also receives B . Induction over the finite cells proves the claim.

Apply this claim to $T = S((n+1)NV)$. Together with (6.5), the criterion of Section 2 proves the theorem. $\square$

7. CLOSED STABILIZER SUBGROUPS

Proof of Theorem 1.1. We choose an open normal uniform pro- p subgroup with solvable quotient. Proposition 3.1 applies to the uniform subgroup, and a composition series of the quotient allows us to apply Theorem 6.1 to the remaining extensions.

Reduction in the maximal order gives an exact sequence

$$1 \longrightarrow 1 + \mathfrak{m}_{\mathcal{O}_n} \longrightarrow \mathbb{G}_n \longrightarrow \mathbf{F}_{p^n}^\times \rtimes \text{Gal}(\mathbf{F}_{p^n}/\mathbf{F}_p) \longrightarrow 1.$$

The kernel is pro- p and the quotient is finite solvable; see [8, Definition 4.2 and Section 4.1] for the reduction map and its compatibility with the Galois action. For any finite quotient of a closed subgroup $G \leq \mathbb{G}_n$, the image of $G \cap (1 + \mathfrak{m}_{\mathcal{O}_n})$ is a normal finite p -group. Quotienting by this image gives a quotient of $G/(G \cap (1 + \mathfrak{m}_{\mathcal{O}_n}))$. Reduction embeds this latter group in the displayed finite solvable group, so the remaining quotient is solvable. Thus every finite quotient of G is solvable.

A closed subgroup $G \leq \mathbb{G}_n$ is compact p -adic analytic. It has an open normal uniform pro- p subgroup by [5, Corollary 8.34]. A sufficiently deep characteristic power subgroup is still uniform

and lies inside $1 + \mathfrak{m}_{\mathcal{O}_n}$; denote it by U . A composition series of the finite solvable group G/U pulls back to a subnormal chain

$$U = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_s = G, \quad G_i/G_{i-1} \cong C_{\ell_i}.$$

For each i , the map $E_n^{hG_i} \rightarrow E_n^{hG_{i-1}}$ is a faithful finite C_{ℓ_i} -Galois extension in $\mathrm{Sp}_{K(n)}$, by [10, Theorem 3.1.12]. It is descendable by [10, Corollary 3.2.10]. Theorem 6.1 therefore applies to each such map. Since U has finite p -cohomological dimension, Proposition 3.1 applies to $E_n^{hU} \rightarrow E_n$. Lemma 2.2 gives its $\mathbb{E}_\infty$ -descendability also in $\mathrm{Sp}_{K(n)}$. Composition proves $\mathbb{E}_\infty$ -descendability of

$$E_n^{hG} \longrightarrow E_n^{hU} \longrightarrow E_n$$

in $\mathrm{Sp}_{K(n)}$. A second application of Lemma 2.2 gives the ordinary assertion. $\square$

Remark 7.1. The finite residue field in the definition of E_n is part of the hypothesis. Replacing it by an algebraic closure changes the Galois factor and requires a separate continuous argument. The theorem also does not claim DAlg-descendability: periodic Morava ring spectra have not been equipped here with structures as derived commutative rings in that sense.

APPENDIX A. A PARTIAL MORPHISM WITHOUT UNIT COMPATIBILITY

The unit-insertion conditions in Lemma 4.1 cannot be replaced by compatibility on the all-unit input. We give a partial nonunital morphism under the scalar algebra that does not extend, although the proposed obstruction formed from the unit cofiber vanishes. The example concerns the chosen partial morphism; its source does admit an algebra augmentation.

Work in rational chain complexes with homological grading, and take tensor products over $H\mathbf{Q}$. Let

$$X = H\mathbf{Q} \oplus \Sigma^{-1}H\mathbf{Q}\varepsilon, \quad |\varepsilon| = -1, \quad \varepsilon^2 = 0,$$

and let $q : X \rightarrow H\mathbf{Q}$ be projection. Both algebras are strictly unital and commutative, with zero differentials.

The positive operad $\varphi_2\mathrm{Comm}$ is freely generated by a symmetric binary operation. A morphism structure on q includes a homotopy between $q\mu_X$ and $\mu_{H\mathbf{Q}}(q \otimes q)$. These maps agree, but choose the nonconstant loop represented by the degree-one map

$$h(1, \varepsilon) = h(\varepsilon, 1) = 1, \quad h(1, 1) = h(\varepsilon, \varepsilon) = 0.$$

This gives a partial nonunital algebra morphism. Its restriction along the scalar algebra is the identity partial morphism, since $h(1, 1) = 0$.

For the two bracketings of three inputs, the induced binary-tree homotopies are

$$\begin{aligned} H_{(xy)z}(x, y, z) &= h(xy, z) + h(x, y)q(z), \\ H_{x(yz)}(x, y, z) &= h(x, yz) + q(x)h(y, z). \end{aligned}$$

On $(1, 1, \varepsilon)$ their values are 1 and 2, respectively. Their difference is a nonzero cycle and cannot be a boundary. The ternary commutative operation identifies these binary trees, so the chosen partial morphism does not extend through arity three.

We now compute the expression that a unit-cofiber obstruction formula would assign to it. Let Std_{Σ_3} be the standard rational representation, the kernel of the sum map $\mathbf{Q}^3 \rightarrow \mathbf{Q}$ with its permutation action. Rationally,

$$C_3 \simeq \Sigma \mathrm{Std}_{\Sigma_3}, \quad \mathrm{cofib}(H\mathbf{Q} \rightarrow X) = \Sigma^{-1}H\mathbf{Q}.$$

The first equivalence follows because the arity-three latching map is the fold from the three binary trees to the single commutative operation. The permutation action on the third tensor power of $\Sigma^{-1}H\mathbf{Q}$ is the sign action. Exactness of rational coinvariants gives

$$(C_3 \otimes \operatorname{cofib}(H\mathbf{Q} \rightarrow X)^{\otimes 3})_{h\Sigma_3} \simeq \Sigma^{-2}(\operatorname{Std}_{\Sigma_3} \otimes \operatorname{sgn})_{\Sigma_3} = 0.$$

Thus an all-unit restriction does not suffice for the unit-cofiber obstruction formula. The individual unit-insertion conditions in Lemma 4.1 exclude this partial morphism.

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